

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mathematics

2. **Subject Name:** Applied Computational Methods **L-T-P:** 3-0-0 **Credits:** 3

3. **Pre-requisites:** NA

4. **Syllabus and reference books:**

Direct and Iterative methods for solving linear systems of equations: QR Decomposition, Thomas algorithms, Basic theory of linear fixed-point iteration, Successive Overrelaxation, Iterative methods for matrix inversion, Krylov subspace methods.

Numerical solutions of PDEs: Numerical solutions of parabolic and elliptic PDEs: finite difference methods, consistency, convergence, stability, Crank-Nicolson method, iterative methods; Numerical solutions of scalar & vector conservation laws: weak solution, Rankine-Hugoniot condition, entropy conditions, Riemann problem, rarefaction waves, shock and contact discontinuities, upwinding, CFL condition, finite difference and finite volume methods, Lax-Friedrichs method, Lax-Wendroff theorem, Godunov's method, Roe's linearization, eigenvalue stability of finite difference methods, consistency and method of modified PDEs, Lax equivalence theorem, TVD schemes, high-resolution schemes, flux and slope limiters, Iterative methods.

Reference Books:

- 1 A. S. Ackleh, E. J Allen, R. B. Kearfott, P. Seshaiyer, Classical and Modern Numerical Analysis, Theory, Methods, and Practice, CRC Press.
2. Joe D. Hoffman, Numerical Methods for Engineers and Scientists, CRC Press
3. Uri M. Ascher and Chen Greif, A first Course in Numerical Methods, Computational Science and Engineering, SIAM.
4. D. Kincaid and W. Cheney, Numerical Analysis: Mathematics of Scientific Computing, AMS.
5. J.S. Hesthaven, Numerical methods for conservation laws, vol. 18 of Computational Science & Engineering, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2018. From analysis to algorithms.
6. R. LeVeque, Finite difference methods for ordinary and partial differential equations. SIAM, 2007.

1. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Direct and Iterative methods for solving linear systems of equations	10
2.	Numerical solutions of Parabolic and Elliptic PDEs	10
3.	Numerical solutions of scalar conservation laws	10
4.	Numerical solutions of vector conservation laws	10
Total number of hours		40